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Mind, Brain, and Meaning

Blind as a Bat, not “Like a Bat”

The intricate relationship between consciousness, perception, and sensation unveils a complex interplay in how we construct our understanding of the world. While consciousness isn't always a requisite for perception, the dynamic synergy between consciousness and perception is evident in the bottom-up and top-down processing of sensory stimuli. Through phenomena like blindsight, where individuals respond to visual stimuli despite lacking conscious awareness, we witness how sensation operates independently of consciousness yet remains intertwined with it. Furthermore, the interpretation of sensory data in visual illusions and olfactory perception underscores the pivotal role of consciousness in shaping our perceptual reality. Therefore, although consciousness is not always a necessary component for perception, consciousness and perception closely work together to employ bottom-up and top-down processing of sensation to construct our individual world views.

To first examine the somewhat independent process of sensation from consciousness, we will discuss blindsight, a neurological phenomenon in which cortically blind individuals are still able to respond to visual stimuli they are unconscious of. Borrowing Paller and Suzuki's distillation of the process of consciousness to “a moment of awareness” that defines a conscious experience and is integral to the broader state of consciousness itself, unconsciousness in blindsight refers to the cortical blindness itself: a lack of awareness of visual stimuli. Individuals who experience blindsight can, for example, detect motion without “seeing” anything. A patient with cortical blindness may not be receiving sensations that allow them to experience “vision” due to the damage in their primary visual cortex, but the visual motion area (V5) still receives sensory input even if the patient is unaware that they are facing a moving object. Sensation is a physical process and will occur if the necessary biological components are intact. It is only after the reception of sensation that, through top-down processing and experiential data, the patient can interpret this sensory data and perceive the motion. In this example, it seems that perception also does not require consciousness. I would argue that perception in blindsight calls upon

prior conscious experiences – or, at least it is hard for me to imagine that a cortically blind individual would label motion as “motion” if they had never labeled the sensation represented by activity in V5 as motion before. Therefore, even in blindsight, consciousness, or more specifically, a conscious experience, greatly guides the perceptual process.

The outcomes of perceptual processes, in both olfactory and visual perception, do not directly follow from sensory information, hinting at the idea that a process, consciousness, directs the manipulation and interpretation of sensory information to produce our perception. Visual illusions demonstrate this well. On slide 7 of lecture 11, we see two flower-like patterns, one with a seemingly small orange dot surrounded by larger grayish blue dots and the other with a relatively large orange dot surrounded by smaller grayish blue dots. When asked the question, which circle is bigger, most of the class recognized that without being exposed to this illusion before, they would have said the orange dot surrounded by small gray circles was bigger. Objectively, the two orange center dots are the same size and only the surrounding gray dots differ in size. Sensationally, we are seeing two orange dots that have the same dimension, but our awareness, or consciousness, of the relatively different-sized dots around the center dot leads us to believe that they are different sizes. We have no reason to believe that the dots should be any specific size if we focus our attention on each dot individually, but because we are aware of the image as a whole, we are conscious of the context that surrounds each center dot, we perceive the dots’ size relatively to its surroundings.

Similarly, the conscious experience of smell helps to explain how humans develop a sense of smell from the non-uniform and unstable process of olfactory perception (Lecture 12, Slides 16-20). Barwich, in their critique of olfactory objects, tells us that, “In olfaction, these mechanisms exhibit higher levels of variation, notably already at the periphery; so much so that the identity of olfactory sensations is shown to be inherently variable in its causal basis and does not comply with the idea of constancy in perceptual objecthood” (Barwich, 2). There is no finite or objective truth for what a certain smell or olfactory sensation should be perceived as, so how can we say we perceive smell at all? From sensory information from the reception of chemical odorants, every person has an individual perception of that

smell. The individual perceptions are differentiated by each person's consciousness of the sensation.

Individuals who have been habituated to a certain experience of an environment may become unconscious of the smell of their childhood home, for example, whereas guests, aware of the novelty of the setting they are in, notice that it smells different than their own home. Or, someone who associates the smell of vanilla with food poisoning may perceive the smell of vanilla as “sickening” while others perceive it as a “sweet” odor. Again, these perceptions are tied to a conscious experience attached to the individual.

Thomas Nagel’s paper, “What Is It Like to Be a Bat?” actually helps us understand this idea: “To imagine something perceptually, we put ourselves in a conscious state resembling the state we would be in if we perceived it” (Nagel, 446). Essentially, Nagel is telling us that consciousness and perception have a directly proportional relationship. Altering our conscious experience/state helps us alter our perception.

Finally, putting Nagel and Tsakiris in conversation with one another solidifies and summarizes consciousness’s guiding role in perception’s interpretation of sensory information. Nagel’s paper also introduces a distinction between human consciousness and a potential bat consciousness. Essentially, Nagel offers the idea that the form of a conscious experience is defined by what it is like to *be* any specific organism (Nagel, 436). Superimposing this idea onto specific bodies within human consciousness, it seems logical that every individual's conscious experience can be defined by what it is like to *be* any individual human in their body. Just as Nagel said it seems impossible for a human to know what it is like for a bat to be a bat, it seems plausible that one cannot know what it is like for another person to be themselves. Tsakiris’ paper on the multisensory self addresses this idea by discussing body ownership, the bodily self, and interoception: “awareness of the interoceptive body may be fundamental to the unity of the self because interoception is required for the experience of a unified, non-hollow self, and to its stability because interoception can counteract the ever-changing influence of exteroceptive signals” (Tsakiris, 604). Mapping this quote to the ideas previously presented in this paper, Tsakiris is touching on the importance of consciousness in the perception of external stimuli – the conscious experience of being oneself stabilizes how we perceive stimuli. Tsakiris goes on to say that the perceptual learning process first involves asserting a personal ownership, “mineness” over a new conscious

experience or “body” before the altered perceptual state can follow (Tsakiris, 604). This is compatible with Nagel’s view that it is impossible to assume another person’s experience because Tsakiris emphasizes the necessity of maintaining a consistent ownership and mineness of the self - a modification of the self’s conscious experience rather than the acquisition of a different self or experience. Overall, the synthesis of these perspectives highlights the significant role of consciousness in shaping perception, elucidating its function in guiding the interpretation of sensory input and ultimately influencing our comprehension of the external world.

Works Cited

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